

Name: _____

Roll # _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR(Computer Science)
SPRING SEMESTER EXAMINATIONS 2026

Batch 2025

Dated : 10-JUN-26

Max Marks : 60

Time : 3 Hours

Logic Design & Switching Theory - CS-251

Note: Attempt all questions.

Q1 (CLO-1 Cognitive Level C2 (Understanding)) [12 Marks]

a. Simplify using Boolean Algebra:

i. $F(W, X, Y, Z) = (WX + W\bar{Y})(W + X) + WX(\bar{X} + \bar{Y})$

ii. $F(A, B, C) = (X + \bar{Y}Z)(\bar{X}YZ)$

iii. $F(A, B, C) = AB + BC(B + C)$

b. In a digital security system, two binary access codes are compared before granting entry. Explain how a 4-bit Magnitude Comparator can be used to determine whether the entered code is greater than, less than, or equal to the stored code.

Q2 (CLO-1 Cognitive Level C2 (Understanding)) [12 Marks]

a. In a hospital emergency system, multiple patients may request assistance at the same time. Explain how an 8-to-3 High-Priority Encoder can be used to identify and process the most critical patient request first.

b. Differentiate between

- Synchronous & Asynchronous counters
- Sequential & Combinational Logic
- Moore & Mealy Machines

Q3 (CLO-2 Cognitive Level C4 (Analysis)) [12 Marks]

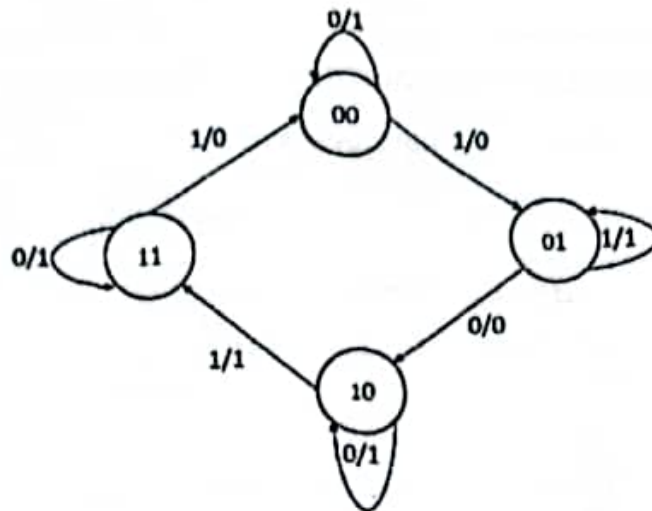
- Demonstrate with the help of a diagram and working table how a 4-bit asynchronous counter can be connected to limit it to form a Mod-10 counter.
- Compare JK flip flop with SR flip flop in terms of characteristic table, equation, excitation table and logic diagram.

Q4 (CLO-2 Cognitive Level C4 (Analysis)) [12 Marks]

- Analyze the operation of a Parallel-In Serial-Out (PISO) shift register with the help of a diagram and explain its modes of operation.
- Analyze the given state diagram and determine whether it represents a Mealy or Moore state machine. Convert the diagram into the other type of state machine and construct the state tables for both machines.

P.T.O

X.YZ



Q5 (CLO-2 Cognitive Level C4 (Analysis)) [12 Marks]

- Analyze the working of a 4-bit Carry Look-Ahead Adder and compare it with a Ripple Carry Adder in terms of speed.
- Design a sequential circuit using T flipflop for the given state diagram.

